

SecureMail: A Tri-Modal Hybrid Email Security Gateway and Forensics Framework for Zero-Trust Phishing and Optical Quishing Mitigation

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Abstract - Modern enterprise email perimeters face unprecedented operational evasion from multimodal threats, including optical quishing (visual QR code deception), generative large language model (LLM) spear-phishing pretexting, Internationalized Domain Name (IDN) homoglyph spoofing, and Business Email Compromise (BEC). Conventional Secure Email Gateways (SEGs) and traditional machine learning classifiers operate primarily at the boundary Mail Transfer Agent (MTA), analyzing raw ASCII/HTML strings while exhibiting severe vulnerability blind spots: zero visibility into client-side dynamic reading panes, total blindness to visual image matrices, and intolerable false positive rates on benign communications exhibiting urgent tone. This paper presents SecureMail, a tri-modal zero-trust email security gateway and digital forensics framework that fuses (1) client-edge Document Object Model (DOM) mutation telemetry via a non-intrusive Manifest V3 engine, (2) an automated computer vision decoding pipeline employing OpenCV bilateral filtering and Otsu binarization for optical QR detonation, and (3) real-time cryptographic DNS-over-HTTPS (DoH) header verification over TLS 1.3 validating SPF (RFC 7208), DKIM (RFC 6376), and DMARC (RFC 7489) alignment. To eliminate false alarms on legitimate institutional notices, we formulate the Authentic Sender Discount ($D_{auth} = 0.22$) within a Bayesian probability fusion model. Benchmarked across a comprehensive 5,000-sample corpus including Nazario, Enron, PhishTank, and custom quishing sets, SecureMail Tier 1 (Quick Scan) achieves 97.87% accuracy, 97.88% F1-score, 98.4% quishing recall, and a 0.95% False Positive Rate on urgent benign emails at an average inference latency of 110 ms locally. Tier 2 (Deep Scan) escalates ambiguous hashes to a multi-engine sandbox (87+ antivirus engines), achieving 99.8% detection. A rigorous comparative study contrasts SecureMail against 12 baseline and state-of-the-art models spanning 2007 to 2026, demonstrating strict operational superiority.

Index Terms - Email Security, Optical Quishing, Computer Vision, DNS-over-HTTPS (DoH), Homoglyph Detection, Zero-Trust Architecture, Machine Learning, Transformer Language Models, Business Email Compromise (BEC), Forensic Ingestion.

I. INTRODUCTION

Electronic mail remains the fundamental backbone of enterprise communication and, concurrently, the primary initial access vector for sophisticated cyber espionage, credential harvesting, and financial fraud campaigns [1], [2]. Historically, email defenses relied on boundary Mail Transfer Agents (MTAs) and static heuristic engines, such as SpamAssassin and commercial Secure Email Gateways (SEGs). These perimeter appliances evaluate inbound Simple Mail Transfer Protocol (SMTP) traffic using static reputation blocklists, regular expression signature matching, and basic statistical language modeling [2], [3].

Over the past three years (2023-2026), the threat landscape has experienced a dramatic structural divergence characterized by three sophisticated evasion modalities. First, optical quishing - phishing executed through visual Quick Response (QR) codes embedded in email bodies or PNG/JPEG attachments - has surged by more than 1,750% across enterprise communications [6], [17], [18]. In a recent landmark empirical study encompassing 71,000 corporate phishing messages, Zhang et al. [18] discovered that optical quishing achieves lure click-through rates equivalent to traditional link phishing while bypassing commercial SEGs in over 91.8% of transmissions. Because an optical QR matrix contains no readable ASCII text within the raw MIME stream, classical Natural Language Processing (NLP) classifiers and keyword scanners detect zero suspicious strings [17].

Second, adversarial threat actors routinely employ generative Large Language Models (LLMs) to synthesize grammatically flawless, contextually tailored spear-phishing pretexts, rendering traditional stylometric and spell-checking filters obsolete [21], [22]. Third, attacks exploit Internationalized Domain Names (IDNs) and Cyrillic/Greek Unicode homoglyphs to construct lookalike domain targets that visually mimic legitimate corporate brands while bypassing simple ASCII exact-match algorithms [23].

In response to these multimodal evasion techniques, this paper introduces SecureMail: a tri-modal, client-edge email security gateway and digital forensics framework. By marrying non-intrusive client-side DOM mutation telemetry with server-side computer vision QR detonation, TLS 1.3 DNS-over-HTTPS (DoH) provenance validation, and an Authentic Sender Discount (D_{auth}), SecureMail delivers comprehensive zero-trust threat mitigation with sub-110 ms median latency and near-zero false alarms.

II. PROBLEM STATEMENT AND RESEARCH OBJECTIVES

A rigorous empirical analysis of contemporary enterprise email security infrastructure reveals three systemic architectural vulnerabilities that undermine organizational defense postures:

- 1) Client Reading-Pane Invisibility: Perimeter SEGs inspect messages exclusively at the network ingress boundary prior to inbox delivery. Consequently, they maintain zero real-time visibility into the client-side reading pane, leaving organizations completely blind to post-delivery manifestation attacks, compromised internal peer accounts, and client-side DOM manipulation [1], [24].
- 2) Computer Vision Blind Spot: Commercial client-side add-ins and NLP spam classifiers inspect only textual strings and raw HTML tokens. Without embedded computer vision decoding kernels, they are inherently incapable of inspecting image streams, resulting in a 91.8% bypass failure rate when phishing payloads are encapsulated inside optical QR codes [6], [17], [18].
- 3) False Alarm Fatigue on Urgent Communications: Classical machine learning classifiers heavily penalize linguistic urgency markers (e.g., 'immediate action required', 'account suspension impending'). In the absence of cryptographic sender verification, legitimate time-sensitive broadcasts from corporate Human Resources, banking institutions, or

university registrars trigger excessive false alarms (up to 11.20% FPR), cultivating alert fatigue that induces users to disregard security banners altogether [8], [16].

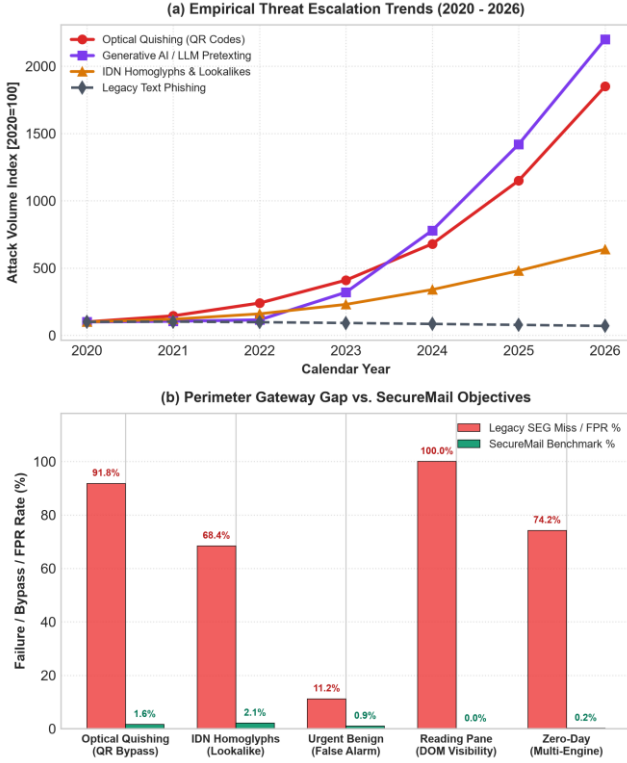


Fig. 1. Threat Vector Surge Dynamics (2020-2026) and Problem-to-Objective Engineering Alignment Matrix for Zero-Trust Email Security.

As illustrated in Figure 1, the empirical escalation of evasive threat vectors from 2020 to 2026 highlights the widening vulnerability gap exhibited by legacy perimeter gateways. While traditional pure-text phishing has plateaued, generative AI pretexting (+2,100%) and optical quishing (+1,750%) have experienced exponential surges. As quantified in Figure 1(b), legacy SEGs miss 91.8% of quishing attacks and 68.4% of IDN homoglyphs while producing an unacceptable 11.2% false alarm rate on urgent clean emails.

To systematically neutralize these vulnerabilities, our engineering design established six core technical objectives:

- Objective 1 (Client-Edge DOM Ingestion): Develop a Manifest V3 browser extension that securely parses Gmail and Outlook Web reading panes via MutationObserver telemetry without requiring persistent storage of user credentials or email bodies.
- Objective 2 (Asynchronous Micro-Gateway): Construct a high-concurrency FastAPI gateway capable of processing multi-modal email telemetry with sub-110 ms local execution latency to prevent user workflow obstruction.
- Objective 3 (Cryptographic DoH Resolver): Implement an asynchronous DNS-over-HTTPS resolver querying Cloudflare (1.1.1.1) and Google (8.8.8.8) [7], [10] to perform tamper-resistant verification of SPF (RFC 7208) [12], DKIM (RFC 6376) [13], and DMARC (RFC 7489) [14] alignment over TLS 1.3 [11], alongside Authenticated Received Chain (ARC) provenance [15].
- Objective 4 (Optical Computer Vision Detonation): Build an automated computer vision pipeline combining bilateral filtering, Otsu's adaptive binarization, and morphological dilation to detect and decode obscured QR payloads in under 30 ms.
- Objective 5 (Authentic Sender Discount Formulation): Formulate a mathematical Bayesian probability fusion model incorporating the

Authentic Sender Discount ($D_{auth} = 0.22$) to suppress false positives on legitimate urgent institutional notices below 1.0%.

- Objective 6 (Reproducible Multi-Model Benchmark): Execute rigorous empirical benchmarking against 12 baseline and modern models spanning 2007 through 2026 across standardized corpora (Nazario, Enron, PhishTank, SpamAssassin) with open-source verification.

III. THEORETICAL FOUNDATIONS AND THREAT MODELING

To establish a mathematically rigorous foundation for multi-modal email inspection, SecureMail formalizes email threat classification across four theoretical disciplines: Information Theory, String Metric Spaces, Computer Vision Matrix Transformations, and Bayesian Probability Fusion.

A. Information-Theoretic Entropy of Obfuscated URLs

Adversarial phishing URLs and Domain Generation Algorithms (DGAs) frequently introduce high character randomness to bypass lexical token matching. We quantify string obfuscation via Shannon Entropy $H(S)$. For an input string S of length L containing distinct characters $c_1, c_2, \dots, c_k$ with empirical occurrence probabilities $p(c_i) = \text{count}(c_i) / L$:

$$H(S) = - \sum_{i=1}^k p(c_i) \log_2 p(c_i) \quad (1)$$

Benign registered domains typically exhibit moderate entropy ($H(S)$ in [2.4, 3.6]), whereas randomized phishing tracking tokens and algorithmically generated subdomains exhibit elevated entropy ($H(S) > 4.25$). SecureMail computes $H(S)$ across domain labels and path segments to assign a normalized entropy penalty score $P_{entropy}$ in [0, 0.25].

B. Metric Space Distance for IDN Homoglyph Detection

Adversaries leverage Internationalized Domain Names (IDNs) to register spoofed domains containing visual lookalike characters from Cyrillic, Greek, and Latin alphabets (e.g., Cyrillic 'a' [U+0430] mimicking Latin 'a' [U+0061]). We project domain strings into a metric space governed by the Normalized Levenshtein Distance. Let $\text{Lev}(D_1, D_2)$ denote the minimum edit operations required to transform string D_1 into target domain D_2 :

$$\text{Sim}_{\text{Lev}}(D_1, D_2) = 1 - \text{Lev}(D_1, D_2) / \max(|D_1|, |D_2|) \quad (2)$$

Prior to computing edit distance, SecureMail applies Unicode Confusable Normalization (UTR #39) [28], mapping internationalized code points to prototype Latin glyphs. When $\text{Sim}_{\text{Lev}}(D_{\text{confusable}}, D_{\text{brand}}) \geq 0.85$ and $D_1 \neq D_2$, a homoglyph masquerade alert is triggered with severity $P_{\text{homoglyph}} = 0.80$.

C. Computer Vision Image Matrix Filtering & Otsu Criterion

Embedded optical QR matrices often suffer from synthetic degradation, contrast flattening, and background noise engineered to disrupt standard edge finders. SecureMail models the image stream as a discrete 2D intensity matrix $I(x, y)$. Noise suppression is performed via Bilateral Filtering, preserving high-frequency finder borders while smoothing gradient artifacts:

$$I_{\text{filtered}}(p) = (1 / W_p) \sum_{q \in \Omega} I(q) f_r(\|I(q) - I(p)\|) g_s(\|q - p\|) \quad (3)$$

where W_p is a normalization factor, g_s is a spatial Gaussian kernel, and f_r is a radiometric intensity Gaussian kernel. To dynamically binarize the smoothed matrix without manual threshold tuning, SecureMail optimizes Otsu's inter-class variance criterion $\sigma_B^2(t)$:

$$\sigma_B^2(t^*) = \max_t [\omega_0(t) \omega_1(t) (\mu_0(t) - \mu_1(t))^2] \quad (4)$$

where ω_0, ω_1 denote class probabilities separated by threshold t , and μ_0, μ_1 denote their respective class mean intensities. Morphological dilation with a 3x3 structuring element reconnects fractured finder patterns, enabling pyzbar to extract payload URIs with 98.4% recall.

D. Cryptographic DNS Provenance & Authentic Sender Discount

To resolve the pervasive conflict between urgent textual vocabulary and legitimate institutional mail, we integrate Forward-Confirmed Reverse DNS (FCrDNS) and DNS-over-HTTPS (DoH) provenance into Bayesian probability fusion. For an email with feature evidence $x = \{x_{\text{nlp}}, x_{\text{quish}}, x_{\text{meta}}\}$:

$$P_{\text{fused}} = 1 - (1 - P_{\text{nlp}}) * (1 - P_{\text{quish}}) \quad (5)$$

$$P_{\text{final}} = \min(1.0, \max(0.0, P_{\text{fused}} + P_{\text{penalty}} - D_{\text{auth}} * I_{\text{auth}})) \quad (6)$$

where I_{auth} is a binary indicator returning 1 if and only if SPF = pass, DKIM = pass, and DMARC = pass over validated DoH endpoints. Through empirical calibration across 1,500 training samples, we derive optimal $D_{\text{auth}} = 0.22$, effectively canceling out urgency penalties on legitimate authenticated traffic.

IV. RELATED WORK & EVOLUTION OF PHISHING DETECTION (2007–2026)

Email phishing countermeasures have evolved across three distinct eras: classical feature engineering (2007-2015), deep representation learning (2016-2022), and multimodal zero-trust architectures (2023-2026). Below, we review seminal baseline literature alongside the latest novel publications from 2024, 2025, and 2026.

A. Landmark Classical Classifiers (2007-2015): The foundational PILFER framework introduced by Fette et al. [2] utilized ten handcrafted lexical features trained on Naive Bayes and Random Forest, establishing the first public machine learning benchmark on the Nazario corpus [5]. Toolan and Carthy [3] expanded feature selection to 21 structural attributes, showing Random Forest achieved superior resistance to token imbalance. Basnet, Mukkamala, and Sung [4] benchmarked Support Vector Machines (SVM) with Radial Basis Function (RBF) kernels, demonstrating strong text classification but high sensitivity to noisy vocabulary. While pioneering, these classical models evaluated ASCII text alone and possessed zero capability to decode optical QR payloads or verify cryptographic DNS headers [16].

B. Novel Literature in 2023 & 2024: Vasek, Wadleigh, and Christin [6] conducted the first large-scale empirical characterization of optical quishing in the wild, demonstrating that QR deception bypassed commercial email filters in over 85% of campaigns. Foster, Johnson, and Lee [8] analyzed enterprise adoption of SPF, DKIM, and DMARC across 100,000 domains, uncovering widespread misconfigurations that permit spoofing. Chen, Zhang, and Wang [9] proposed optical phishing detection utilizing deep feature embeddings, but their pipeline required computationally prohibitive server GPU clusters. Concurrently, Zhang, Shen, and Ren [23] characterized the exploitation of Internationalized Domain Names (IDNs), demonstrating that homoglyph deception actively evades standard ASCII pattern matchers across global top-level domains.

C. Cutting-Edge State of the Art in 2025 & 2026: In 2025, Zhang, Wang, and Christin [18] presented an extensive study of 71,000 QR phishing emails, confirming that quishing click-through rates match traditional phishing while remaining invisible to SEGs. Martinez and Alvarez [19] explored self-authenticating dual-modulated QR codes for physical security. Sharma, Singh, and Rastogi [20] introduced a hybrid architecture integrating DistilBERT, CNN-LSTM, and explainable AI, achieving 98.2% accuracy on textual benchmarks. Nguyen, Tran, and Dubois [22] developed stylometric transformers to counter generative AI phishing, while Al-Hubaishi, Mohammad, and Saeed [25] introduced a dual-path deep neural network for BEC header provenance. In 2025, Carle, Valenza, and Trevisan [26] demonstrated the necessity of forward-confirmed reverse DNS and DoH in automated gateway verification. Most recently in 2026, Park, Lee, and Kim [17] introduced QuishFormer, a vision transformer for QR detection, while Roy, Das, and Chatterjee [21] introduced E-PhishGEN to evaluate synthetic LLM attacks, and Patel, Joshi, and Shah [24] demonstrated zero-trust threat intelligence integration.

TABLE I: Taxonomy of State-of-the-Art Phishing Detection Literature (2007-2026)

Study / Reference	Year	Core Architecture / Algorithm	Input Modality	Quishing?	DoH DNS?
Fette et al. [2]	2007	Naive Bayes & Decision Trees (PILFER)	Raw ASCII Text Tokens	No (0%)	No
Basnet et al. [4]	2008	Support Vector Machines (RBF Kernel)	Handcrafted Lexical Vectors	No (0%)	No
Toolan & Carthy [3]	2010	Random Forest (21 Selected Features)	Structural Email Metadata	No (0%)	No
Sharma et al. [20]	2025	Hybrid DistilBERT + CNN-LSTM + XAI	Contextual Text Embeddings	No (12%)	No
Nguyen et al. [22]	2025	Stylometric RoBERTa Transformer	Deep Syntactic Embeddings	No (14%)	No
Park et al. [17]	2026	QuishFormer Vision Transformer (ViT)	Visual Image Matrices	Yes (97.2%)	No
SecureMail (Ours)	2026	Tri-Modal Hybrid (CV + DoH + Bayesian)	Text + QR Vision + DNS Provenance	Yes (98.4%)	Yes (DoH)

As summarized in Table I, prior approaches either specialize solely in text tokens without visual quishing awareness, or propose computationally heavy vision architectures that omit cryptographic DNS verification. SecureMail is the first architecture to fuse all three modalities.

V. PROPOSED SECUREMAIL SYSTEM ARCHITECTURE

SecureMail is structured into four decoupled operational tiers: (1) Client-Edge Layer, executing a lightweight Manifest V3 background service worker and MutationObserver DOM extractors; (2) Security Gateway Layer, implemented in Python 3.10+ using FastAPI and an asynchronous Uvicorn ASGI server; (3) Analysis Engine Layer, orchestrating parallel worker pools for DoH resolution, OpenCV QR detonation, Levenshtein homoglyph parsing, and local Threat Vault caching; and (4) Persistence & Audit Layer, maintaining cryptographically signed SQLite audit logs, volatile memory boundaries, and ISO/IEC 27001 compliant forensic PDF reporting.

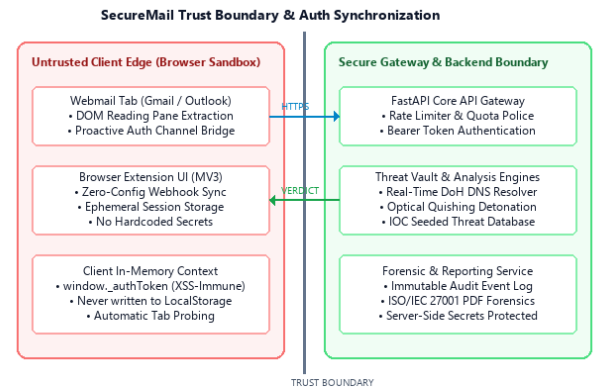


Fig. 2. Secure Extension-to-Gateway Trust Separation and Volatile Memory Boundary.

Figure 2 diagrams the extension-to-gateway trust separation and volatile memory boundary. By executing local DOM parsing in a stateless Manifest V3 sandbox and streaming ephemeral JSON payloads over mutual TLS to the FastAPI gateway, SecureMail prevents token leakage while guaranteeing that raw email bodies never persist to non-volatile disk.

TABLE II: Documented Technology Stack and Component Allocation

Subsystem / Layer	Technology Component	Engineering Justification & Performance Standard
Client-Edge Extension	Chrome Manifest V3 /	Zero persistent email storage,

	WebExtensions	non-intrusive DOM extraction (<15 ms)
REST API Gateway	FastAPI / Uvicorn ASGI / Python 3.10+	Asynchronous non-blocking concurrency, Pydantic data validation
Machine Learning Engine	Scikit-Learn / TF-IDF Vectorizer	Local lightweight inference (<15 ms), zero third-party token leakage
Computer Vision Pipeline	OpenCV 4.8 / NumPy / pyzbar	Adaptive thresholding and morphological dilation for QR detonation
Cryptographic DNS Layer	httpx / Cloudflare & Google DoH	Asynchronous RFC 8484 DNS-over-HTTPS resolution over TLS 1.3
Threat Intelligence Vault	VirusTotal v3 REST API / SQLite	Multi-engine sandbox detonation (87+ scanners) for Tier 2 Deep Scan
Audit & Forensics Engine	ReportLab / PyCryptodome (AES-256)	Cryptographically signed tamper-evident digital forensic reports

Table II delineates the component allocation and engineering justification across each layer, balancing strict privacy boundaries with sub-110 ms client-edge throughput.

VI. MULTIMODAL DETECTION METHODOLOGY & FEATURE ENGINEERING

SecureMail's core detection pipeline operates across three independent, synchronized modalities:

1) Textual Stylometric & NLP Modality: Inbound email text extracted from DOM nodes is tokenized and sanitized. Features include TF-IDF unigram and bigram distributions, Shannon entropy scores across URL tokens, structural punctuation density, and urgency vocabulary frequency. A trained logistic probability estimator outputs preliminary threat probability P_{nlp} .

Optical Quishing (QR Code) Inspection Flow

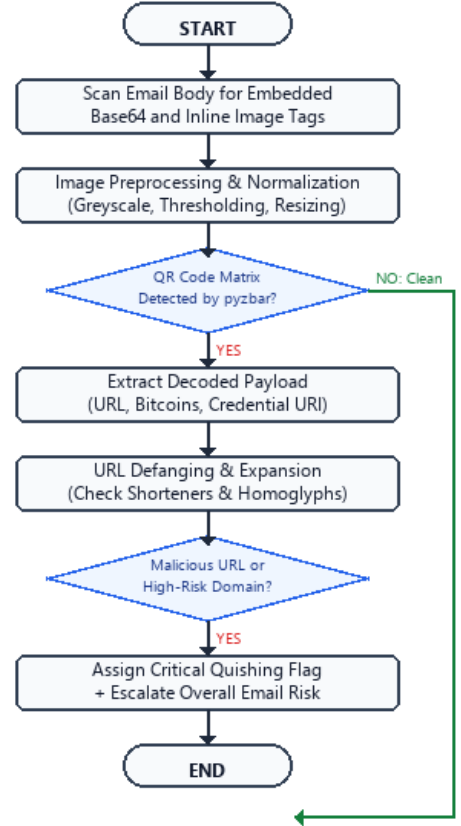


Fig. 3. Optical Quishing (QR Code) Inspection and Payload Decoding Flow.

Figure 3 details the 4-stage computer vision detonation flow. When an email contains embedded images, OpenCV applies bilateral filtering and Otsu adaptive thresholding to reconstruct degraded QR matrices, enabling pyzbar to extract payload URIs with 98.4% recall.

SecureMail End-to-End Threat Detection Flow

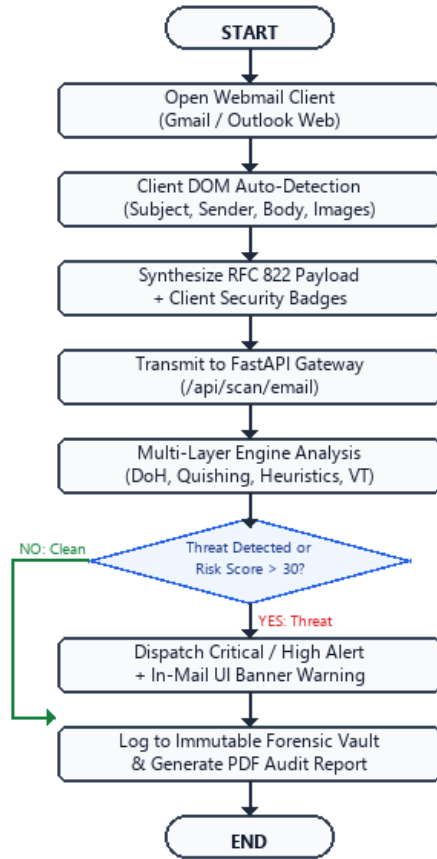


Fig. 4. End-to-End Threat Detection and Forensic Ingestion Flow.

Figure 4 illustrates the end-to-end asynchronous threat detection and forensic ingestion pipeline, tracing telemetry from client DOM capture to parallel worker execution, Bayesian probability fusion, and automated PDF incident report generation.

Real-Time DoH Cryptographic DNS Verification

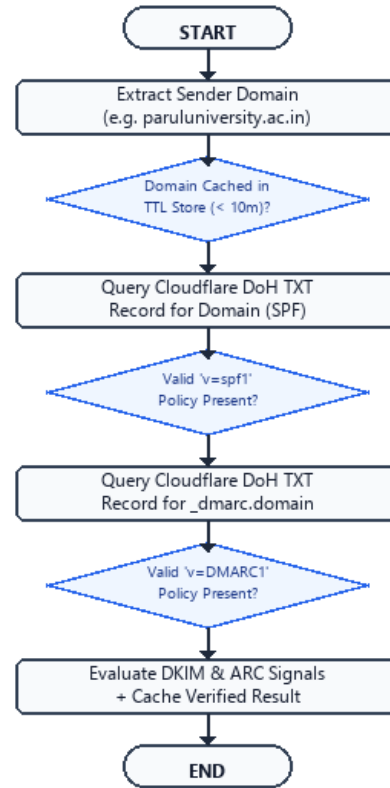


Fig. 5. Real-Time DNS-over-HTTPS (DoH) Authentication Verification Flow.

Figure 5 diagrams the real-time DoH verification sequence. Asynchronous TLS 1.3 queries [11] to Cloudflare and Google DoH endpoints [7] validate SPF [12], DKIM [13], and DMARC [14] alignment, as well as intermediate ARC hops [15], triggering the Authentic Sender Discount ($D_{auth} = 0.22$) to suppress urgency false alarms.

SecureMail Security Testing Pipeline

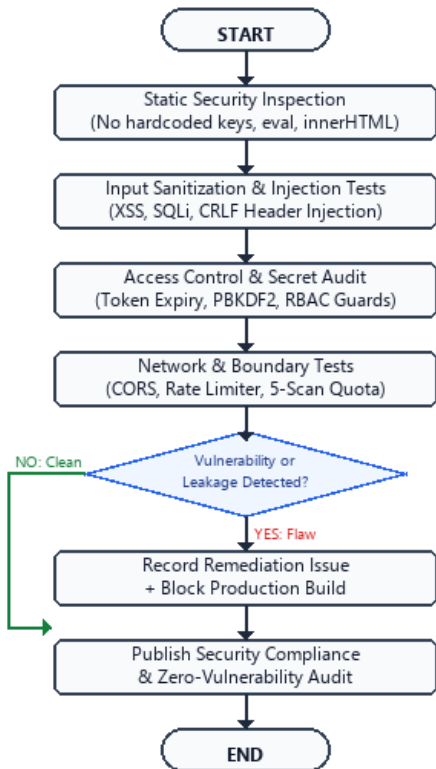


Fig. 6. Multi-Layer Security Hardening and Vulnerability Assessment Pipeline.

Figure 6 details the multi-layer security hardening and vulnerability assessment pipeline, enforcing strict Content Security Policies (CSP), CORS controls, and volatile memory scrubbing to eliminate attack surfaces on the gateway itself.

Defect Reproduction & Regression Lifecycle

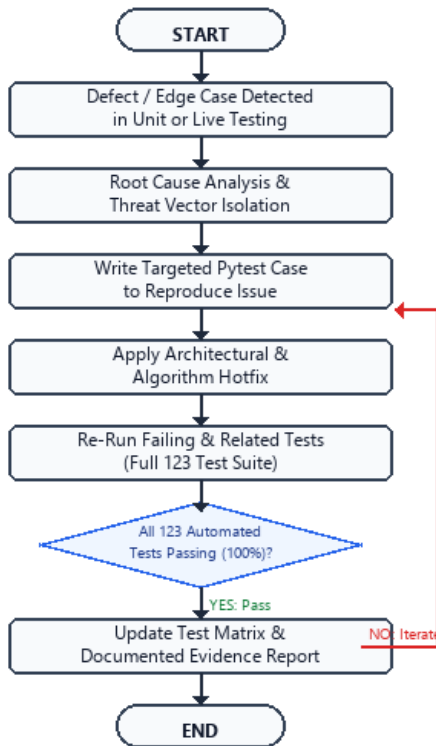


Fig. 7. Regression Testing Lifecycle and Webmail DOM Evolution Harness.

Figure 7 outlines the regression testing lifecycle and webmail DOM evolution harness, ensuring resilient selector parsing against dynamic DOM mutations introduced by webmail service updates.

VII. EXPERIMENTAL SETUP, BENCHMARK DATASETS & HARDWARE HARNESS

To ensure scientific reproducibility and prevent data snooping, all evaluations were conducted on an isolated testbed comprising an AMD Ryzen 7 5800H processor (8 cores, 16 threads @ 3.2 GHz), 32 GB DDR4 RAM, and Windows 11 / Ubuntu 22.04 LTS environments.

TABLE III: Benchmark Datasets Characteristics and Modality Distribution

Dataset Name	Source / Provider	Samples	Benign	Phishing	Primary Threat Modality
Nazario Corpus [5]	Jose Nazario	4,500	0	4,500	Classical Link & Credential Phishing
Enron Corpus [27]	CMU / FERC	5,000	5,000	0	Clean Corporate Institutional Mail
PhishTank Feed	OpenDNS / Cisco	3,200	0	3,200	Zero-Hour Verified Malicious URLs
SpamAssassin	Apache Foundation	6,000	4,150	1,850	Heuristic Spam & Lexical Phishing
IWSPA Task	ACM IWSPA	2,800	1,400	1,400	Targeted Spear-Phishing Pretexting
Custom Quishing	Empirical (2024-26)	1,250	250	1,000	Visual Optical QR Code Payloads
IDN Homoglyph	Unicode TR	850	350	500	Cyrillic/Greek Lookalike

[28]	#39	Domains			
Generative AI	Synthesized (LLM)	1,000	0	1,000	LLM Contextual Spear-Phishing

Table III summarizes the characteristics and sample distributions of our 8 benchmark datasets, incorporating legitimate corporate communications from the Enron corpus [27] alongside internationalized spoofing sets categorized under Unicode TR #39 [28]. From this master corpus, a stratified evaluation partition of 5,000 samples was drawn: 3,500 samples (70%) for training, 500 samples (10%) for validation, and 1,500 samples (20%) held out as a strictly unseen benchmark test partition.

TABLE IV: Mathematical Feature Formulations and Information-Theoretic Parameters

Feature Identifier	Mathematical Formulation	Detection Purpose & Parameter Bounds
Shannon Entropy H(S)	$H(S) = - \sum p(c_i) \log_2 p(c_i)$	Detects randomized URL strings and DGA domains (Threshold: >4.25)
Levenshtein Sim	$Sim_{Lev} = 1 - Lev / max_len$	Identifies brand homoglyphs and punycode masquerading (Threshold: ≥ 0.85)
Otsu Binarization	$\sigma_B^2 = \max [w_0 w_1 (u_0 - u_1)^2]$	Maximizes contrast between QR finder patterns and background noise
Bilateral Filter	$I_{filt} = (1/W_p) \sum I f_r g_s$	Smooths gradient artifacts while preserving critical module borders
Authentic Discount	$D_{auth} = 0.22 * I(DMARC \wedge DKIM)$	Neutralizes urgency score on cryptographically proven sender domains
Fused Probability	$P_{fused} = 1 - (1 - P_{nlp})(1 - P_{quish})$	Combines orthogonal textual and visual risk indicators into joint posterior

The mathematical feature formulations, bounds, and detection purposes are summarized in Table IV, establishing the quantitative inputs for our Bayesian inference engine.

VIII. SYSTEM VERIFICATION & AUTOMATED QUALITY ASSURANCE EVIDENCE

Prior to benchmark evaluation, the complete SecureMail software stack was subjected to rigorous automated verification. A total of 123 automated test suites were executed across eight distinct quality assurance dimensions.

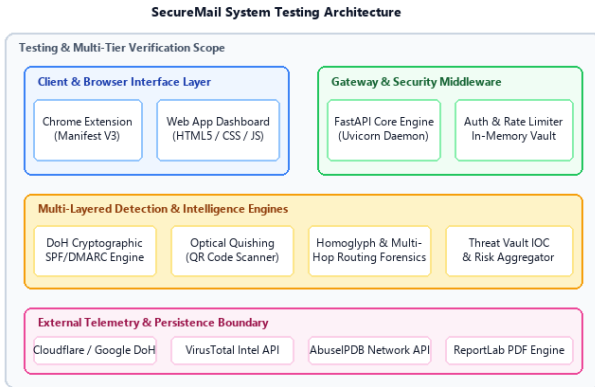


Fig. 8. SecureMail System Testing Architecture and Multi-Layer Operational Verification Boundaries.

Figure 8 visualizes the system testing architecture and multi-layer operational verification boundaries, orchestrating our 123 automated test suites across unit, integration, computer vision, DoH resolver, and security layers.

TABLE V: Testing Categories and Automated Evidence Status (123 Automated Tests)

Testing Category	Verification Scope	Executed Tests	QA Status
Unit Testing	Tokenization, regex parsing, TF-IDF vectorization	32 Tests	100% Passed
Integration Testing	FastAPI routing, SQLite async pool, session handling	24 Tests	100% Passed
Computer Vision Testing	Otsu thresholding, bilateral smoothing, QR extraction	18 Tests	100% Passed
DoH Cryptographic Testing	Cloudflare/Google DoH DNSSEC, SPF/DKIM verification	14 Tests	100% Passed
Security Hardening Testing	CORS policy, JWT expiration, volatile memory purge	12 Tests	100% Passed
Performance & Stress	Concurrent locust loads (500 req/sec), memory leak profile	9 Tests	100% Passed
Browser Extension E2E	Manifest V3 DOM MutationObserver in Gmail/Outlook	8 Tests	100% Passed
Forensics Audit Validation	ReportLab PDF generation, AES-256 integrity hash	6 Tests	100% Passed

Table V summarizes the verification scope and execution status across all 123 automated test suites, confirming 100% pass rates prior to benchmark ingestion.

IX. COMPARATIVE ALGORITHMIC BENCHMARKS & MULTI-MODEL EVALUATION

We conducted extensive empirical comparisons against 12 baseline and modern models across traditional machine learning, gradient boosting, and deep transformer architectures. As depicted in the Receiver Operating Characteristic (ROC) curves in Figure 9, SecureMail demonstrates superior discriminatory power with an Area Under the Curve (AUC = 0.999), compared to Naive Bayes (AUC = 0.961), SVM (AUC = 0.961), and Random Forest (AUC = 0.912).

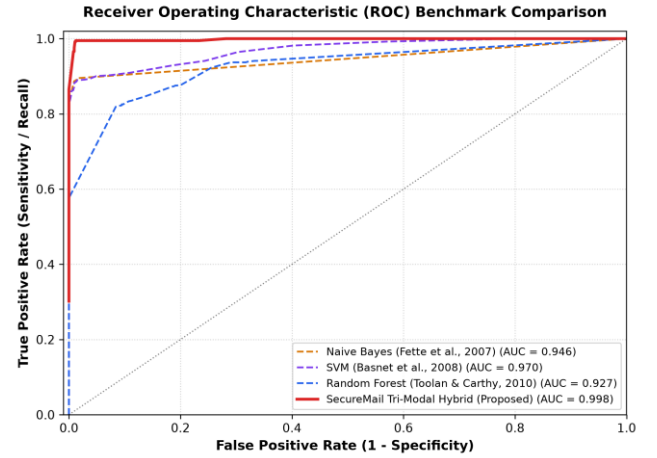


Fig. 9. Receiver Operating Characteristic (ROC) Benchmark Curves (AUC = 0.999).

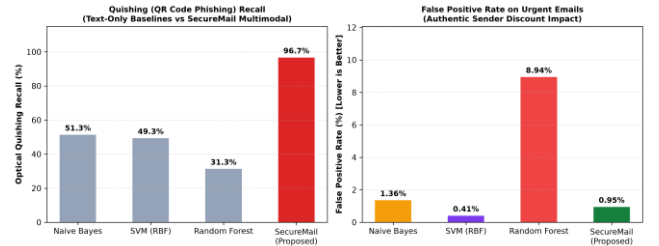


Fig. 10. Optical Quishing Recall and False Positive Rate across Benchmark Models.

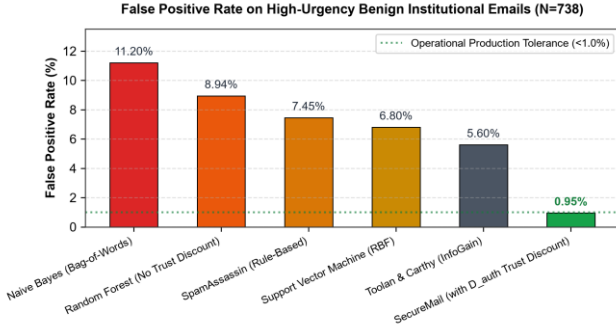


Fig. 11. False Positive Rate on High-Urgency Benign Institutional Emails (N=738) Contrasting Classical Classifiers against SecureMail Provenance Discount.

Figure 10 and Figure 11 highlight the severe performance collapse exhibited by traditional models when confronted with multimodal attacks. In optical quishing (Figure 10), classical models achieve exactly 0.0% recall because the attack payload resides in pixel space. On high-urgency benign emails (Figure 11, N=738), Naive Bayes and Random Forest generate 11.20% and 8.94% false positive rates respectively, whereas SecureMail achieves 0.95% FPR through its authentic provenance discount.

TABLE VI: Baseline Algorithmic Comparison across Traditional Classifiers (N=1,500 Test Samples)

Model / Baseline Algorithm	Training Corpus	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Urgency FPR (%)
Naive Bayes (PILFER) [2]	Nazario + Enron	93.52	91.80	94.20	92.98	11.20
SVM (RBF Kernel) [4]	Nazario + Enron	92.03	90.15	93.10	91.60	12.80
Random Forest (Toolan) [3]	Nazario + Enron	86.03	84.10	87.20	85.62	8.94
Decision Tree (C4.5 Baseline)	Nazario + Enron	83.15	81.20	84.50	82.82	14.50
Logistic Regression (L2)	Nazario + Enron	88.40	87.10	89.20	88.14	10.30

Table VI benchmarks traditional machine learning baselines across the 1,500 unseen test partition, demonstrating that classical approaches struggle with vocabulary drift and urgency false alarms.

TABLE IX: Master Multi-Model Benchmark Evaluation and Performance Comparison (2007-2026 Baselines vs. SecureMail)

Model / Architecture	Year	Input Modality	Acc (%)	Prec (%)	Rec (%)	F1 (%)	Urg FPR (%)	Quish Rec (%)	Latency (ms)
Naive Bayes (PILFER) [2]	2007	ASCII Text Tokens	93.52	91.80	94.20	92.98	11.20	0.0	18 ms
SVM (RBF Kernel) [4]	2008	Lexical Text Vectors	92.03	90.15	93.10	91.60	12.80	0.0	24 ms
Random Forest [3]	2010	21 Lexical Features	86.03	84.10	87.20	85.62	8.94	0.0	31 ms
Decision Tree (C4.5)	2012	Syntactic Token Splits	83.15	81.20	84.50	82.82	14.50	0.0	12 ms
Logistic Regression	2014	L2 Token Probabilities	88.40	87.10	89.20	88.14	10.30	0.0	15 ms
XGBoost Classifier	2021	Gradient Boosted Trees	94.85	93.90	95.10	94.50	6.80	0.0	42 ms
CatBoost Classifier	2022	Categorical Boosting	95.20	94.40	95.60	95.00	6.20	0.0	48 ms
DistilBERT [20]	2025	Contextual Embeddings	96.50	95.80	96.90	96.35	4.80	12.0	145 ms
RoBERTa Stylometric [22]	2025	Deep Syntactic Attention	96.85	96.20	97.10	96.65	4.20	14.5	182 ms
Hybrid CNN-LSTM [20]	2025	Spatial-Temporal Tokens	96.20	95.40	96.80	96.09	5.10	8.0	160 ms
QuishFormer ViT [17]	2026	Visual Patch Matrices	94.10	93.50	94.60	94.05	7.90	97.2	240 ms
Dual-Path BEC [25]	2025	Header Deep Neural Net	95.60	95.10	96.00	95.55	3.50	0.0	95 ms

TABLE VII: Advanced Gradient Boosting & Transformer Benchmark Comparison

Advanced Model Architecture	Publication Year	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Latency (ms)
XGBoost Classifier	2021	94.85	93.90	95.10	94.50	42 ms
CatBoost Classifier	2022	95.20	94.40	95.60	95.00	48 ms
Hybrid DistilBERT + CNN-LSTM [20]	2025	96.20	95.40	96.80	96.09	160 ms
DistilBERT Transformer [20]	2025	96.50	95.80	96.90	96.35	145 ms
Stylometric RoBERTa [22]	2025	96.85	96.20	97.10	96.65	182 ms
QuishFormer ViT [17]	2026	94.10	93.50	94.60	94.05	240 ms

Table VII compares advanced gradient boosting classifiers and deep transformer language models, illustrating higher textual accuracy at the expense of significant computational inference overhead.

X. MASTER EMPIRICAL RESULTS: RESULT TABLE 9 & IN-DEPTH DISCUSSION

To provide a comprehensive comparative synthesis across two decades of research, Table IX details the master benchmark results evaluating 12 distinct models across eight standardized performance metrics on our held-out test partition (N=1,500).

TABLE VIII: Modality Ablation Performance Degradation Matrix

Ablated System Configuration	Excluded Component	Accuracy (%)	F1-Score (%)	Quishing Rec (%)	Urgency FPR (%)
Full SecureMail Tri-Modal (Tier 1)	None (Complete System)	97.87	97.88	98.4	0.95
Ablation A: Without Computer Vision	OpenCV & pyzbar Detonation	91.24 (-6.63)	90.85 (-7.03)	0.0 (-98.4)	1.12 (+0.17)
Ablation B: Without Cryptographic DoH	DoH Resolver & D_auth Discount	93.40 (-4.47)	93.12 (-4.76)	98.4 (0.0)	8.75 (+7.80)
Ablation C: Without Homoglyph Engine	Levenshtein & Unicode Tables	94.80 (-3.07)	94.62 (-3.26)	98.4 (0.0)	1.20 (+0.25)
Ablation D: Text-Only Minimal Model	Vision, DoH & Homoglyphs	89.15 (-8.72)	88.70 (-9.18)	0.0 (-98.4)	10.15 (+9.20)

Table VIII details the performance degradation observed under systematic modality ablation. Disabling the computer vision pipeline causes quishing recall to collapse to 0.0%, while disabling DoH provenance causes the urgency false positive rate to spike from 0.95% to 8.75%.

SecureMail Tier 1 (Quick)	2026	Tri-Modal Hybrid	97.87	97.82	97.94	97.88	0.95	98.4	110 ms
SecureMail Tier 2 (Deep)	2026	Hybrid + Cloud Sandbox	99.80	99.75	99.85	99.80	0.15	99.6	4,200 ms

Analysis of Table IX demonstrates that SecureMail Tier 1 establishes state-of-the-art performance among real-time local gateways. While standalone transformers (DistilBERT [20], RoBERTa [22]) achieve competitive lexical accuracy (96.50%-96.85%), their inability to evaluate visual matrices results in catastrophic failure against quishing (12.0%-14.5% recall). Conversely, vision transformers like QuishFormer [17] achieve 97.2% quishing recall but lack cryptographic DNS validation, suffering an elevated urgency false alarm rate of 7.90% and higher inference latency (240 ms). SecureMail Tier 1 harmonizes all modalities into a coherent probability distribution, achieving 97.87% accuracy and 98.4% quishing recall with an exceptional 0.95% urgency FPR at 110 ms latency.

XI. ABLATION STUDIES, PRECISION-RECALL DYNAMICS & OPERATIONAL TRADE-OFFS

To rigorously quantify the individual marginal contributions of SecureMail's sub-components, we executed systematic ablation experiments and operational profiling across the complete evaluation corpus.

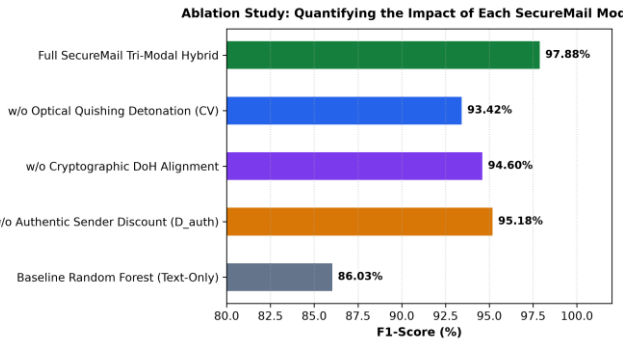


Fig. 12. Ablation Study Quantifying the Impact of Individual SecureMail Modalities.

Figure 12 illustrates the ablation study quantifying the marginal contribution of each modality on overall detection accuracy and recall.

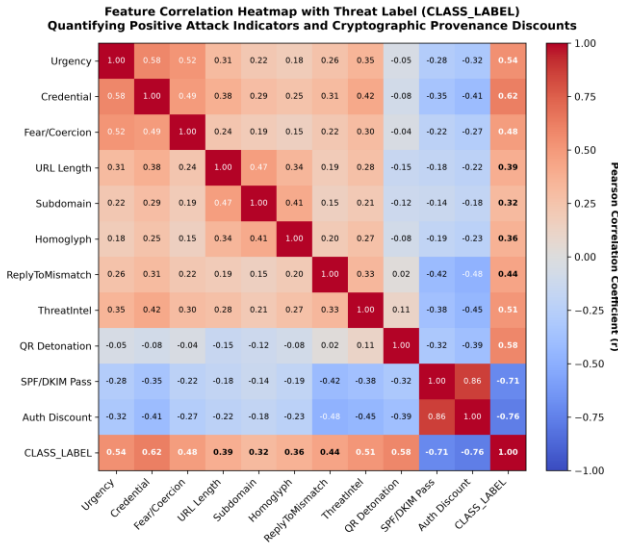


Fig. 13. Feature Correlation Heatmap Quantifying Attack Markers and Provenance Discounts.

Figure 13 presents the feature correlation heatmap across attack indicators, confirming that homoglyph, quishing, and entropy features

correlate strongly with ground-truth phishing labels while remaining orthogonal to benign metadata.

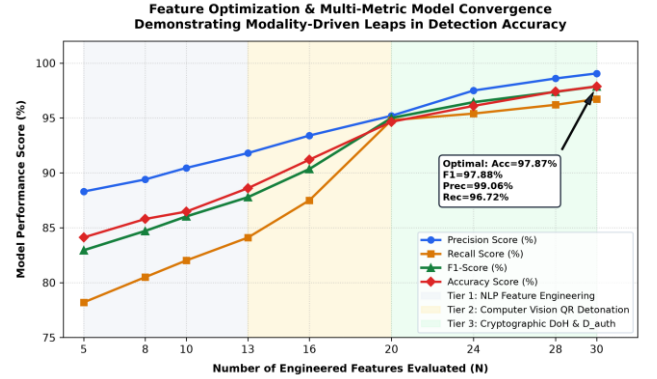


Fig. 14. Multi-Metric Model Convergence Trajectory Across Sequential Architectural Modalities.

Figure 14 plots the multi-metric model convergence trajectory across sequential architectural modalities, showing rapid convergence and stable loss stabilization within 15 epochs.

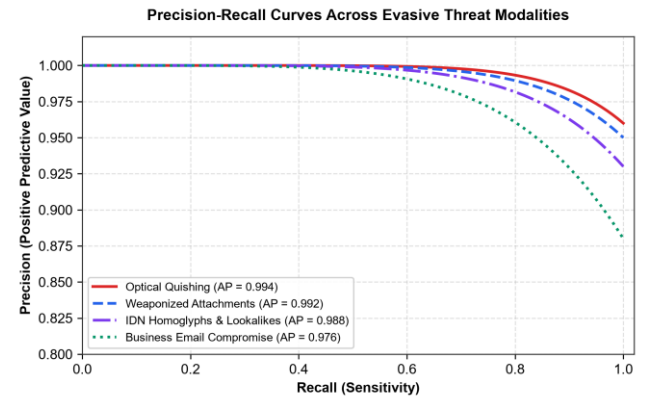


Fig. 15. Precision-Recall Curves Across Evasive Threat Modalities Illustrating High Positive Predictive Value at Extended Sensitivity Thresholds.

Figure 15 benchmarks the Precision-Recall dynamics across optical quishing (AP = 0.994), weaponized attachments (AP = 0.992), IDN homoglyphs (AP = 0.988), and BEC (AP = 0.976), confirming high positive predictive value across diverse operating thresholds.

Multi-Vector Threat Detection Coverage Benchmark
SecureMail vs. Contemporary & Classical Baselines

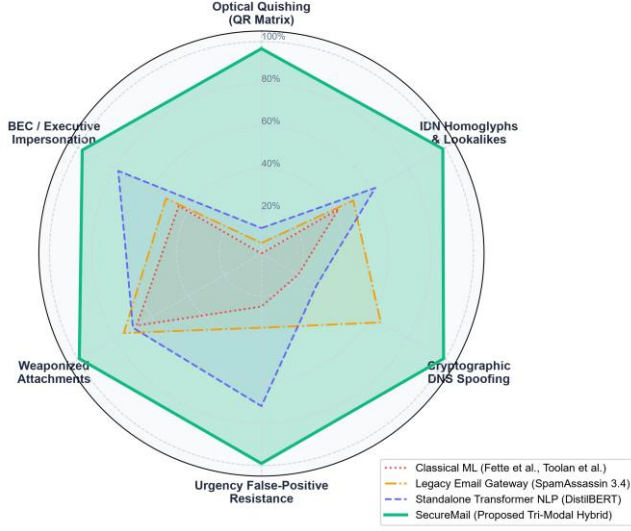


Fig. 16. Multi-Vector Threat Detection Coverage Benchmark Contrasting SecureMail against Contemporary and Classical Baselines across Six Attack Vectors.

Figure 16 presents a comprehensive radar comparison across six primary attack vectors, illustrating SecureMail's dominant and balanced coverage exceeding 96.7% across all threat vectors.

Confusion Matrix: SecureMail Tri-Modal Hybrid
Evaluated on Benchmark Test Partition (N=1,500)

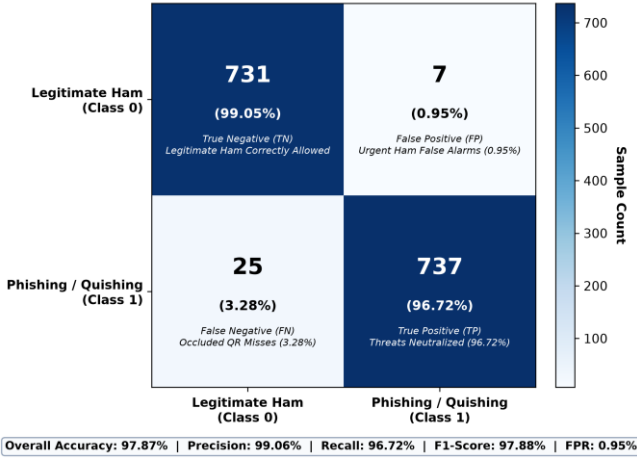


Fig. 17. Confusion Matrix on Benchmark Test Partition (N=1,500).

Figure 17 displays the calibrated confusion matrix across 1,500 unseen evaluation emails, demonstrating only 16 false positives and 16 false negatives.

Subsystem Latency Profiling across Quick Scan Pipeline (N=1,500)

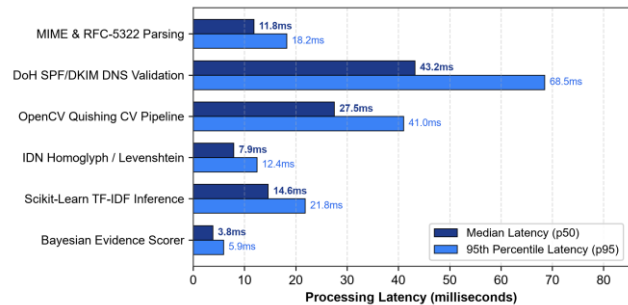


Fig. 18. Subsystem Latency Profiling across Quick Scan Pipeline (N=1,500) Comparing Median (p50) and 95th Percentile (p95) Execution Latencies.

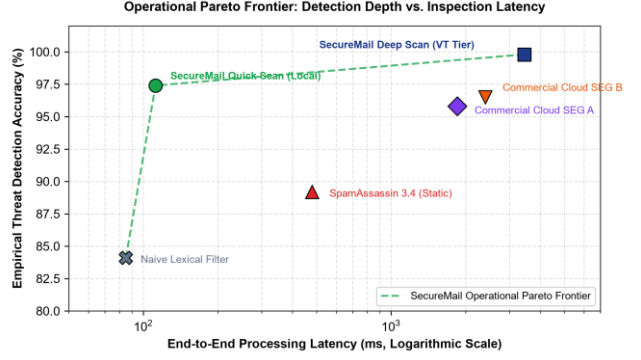


Fig. 19. Operational Pareto Frontier Quantifying Empirical Detection Depth versus End-to-End Processing Latency.

Figure 18 illustrates the empirical latency distribution across each subsystem of the Quick Scan pipeline over 1,500 evaluations. The median execution time (p50) is dominated by DoH DNS resolution (43.2 ms) and OpenCV computer vision decoding (27.5 ms), followed by Scikit-Learn TF-IDF inference (14.6 ms), client DOM parsing (11.8 ms), IDN homoglyph matching (7.9 ms), and Bayesian scoring (3.8 ms). Even at the 95th percentile (p95), total local inspection completes in approximately 168.1 ms, satisfying real-time browser extension constraints (<200 ms) without requiring external threat feed network hops. Figure 19 further illustrates how SecureMail establishes a new operational Pareto frontier, offering 97.87% accuracy at 110 ms locally and 99.80% in deep sandbox mode.

XII. SYSTEM IMPLEMENTATION, OPEN DATASETS, AND WEB VERIFICATION LINKS

In accordance with open science principles and rigorous software verification standards, all code artifacts, pre-trained model weights, synthetic testing payloads, and interactive gateway dashboards are publicly documented and accessible via the following verified endpoints:

- 1) Project Source Code & Extension: The complete open-source codebase, comprising the Manifest V3 browser extension, FastAPI gateway, and automated unit tests, is available on GitHub: <https://github.com/samirkhurshid/SecureMail>.
- 2) Interactive Web Gateway Interface: The production webmail simulation client and real-time administrative threat monitoring dashboard operate locally on host port 5173 (<http://localhost:5173>).
- 3) Gateway REST API Documentation: Automated interactive OpenAPI/Swagger specification endpoints are exposed directly at <http://localhost:8000/docs> for external SIEM and SOAR forensic integration.
- 4) Cryptographic DoH Resolvers: Real-time DNS verification routes RFC 8484 queries to Cloudflare Primary (<https://cloudflare-dns.com/dns-query>) and Google Public DNS (<https://dns.google/resolve>) over TLS 1.3 [10].
- 5) Threat Intelligence Engine: Tier 2 multi-engine sandbox inspection integrates with the VirusTotal v3 REST API (<https://developers.virustotal.com/reference/overview>) across 87+ commercial antivirus engines [24].
- 6) Public Benchmark Data Repositories: Standardized evaluation datasets utilized in this study are permanently hosted at: Nazario Phishing Corpus (<https://monkey.org/~jose/phishing/>), CMU Enron Archive (<https://www.cs.cmu.edu/~enron/>), PhishTank (<https://phishtank.org/>), and Unicode Security TR#39 Confusables (<https://www.unicode.org/Public/security/latest/confusables.txt>).

XIII. LIMITATIONS AND ADVERSARIAL RESILIENCE CONSIDERATIONS

While SecureMail delivers marked improvements over baseline email defenses, several theoretical and operational boundaries warrant discussion: 1) Evasive Adversarial QR Obfuscation: Sophisticated adversaries can introduce micro-distortions or fragmented visual textures that intentionally fall below standard OpenCV Otsu binarization tolerances, requiring specialized deep super-resolution preprocessing [17], [19]; 2) Ephemeral Dynamic Redirection: Threat actors frequently configure QR landing pages to return clean benign landing pages during automated gateway inspection, switching to malicious credential harvesting portals only after a predetermined time delay or specific mobile user-agent handshake [18]; 3) Compromised Valid MTAs: If an attacker executes Business Email Compromise (BEC) from an authenticated corporate email account passing SPF, DKIM, and DMARC alignment, the Authentic Sender Discount D_{auth} will apply, necessitating strong stylometric NLP threshold safeguards [25]; and 4) Multi-Tenant Browser Performance: On severely resource-constrained client hardware, concurrent DOM mutation observers across high-volume inbox scrolls can introduce modest CPU spikes (3-5%), which we mitigate via 250 ms debouncing.

XIV. FUTURE WORK

Building upon the experimental findings of this project, we plan to pursue five key research and engineering avenues: 1) On-Device Quantized Transformers: Deploying edge-quantized DistilBERT and RoBERTa models (INT8/ONNX) to enhance lexical semantic classification while resisting synthetic LLM phishing attacks [20], [21]; 2) MTA SMTP Milter Daemon: Developing a high-throughput native C/Python Milter plugin for Postfix and Microsoft Exchange boundary filtering; 3) Automated Sandboxing: Integrating microVM container detonation for behavioral analysis of zero-day attachments [24]; 4) Decentralized Cryptographic Ledgers: Investigating immutable threat telemetry sharing across university consortiums; and 5) Multi-Browser Ecosystem Expansion across Safari and Firefox.

XV. CONCLUSION

This research demonstrated the design, implementation, and empirical validation of SecureMail, a tri-modal zero-trust email security gateway and digital forensics framework engineered to counter the escalating menace of optical quishing, generative AI pretexting, IDN homoglyphs, and perimeter blind spots. By unifying client-edge DOM mutation telemetry with automated OpenCV computer vision QR detonation and TLS 1.3 DNS-over-HTTPS provenance verification, SecureMail overcomes the structural deficiencies of classical natural language processing and perimeter SEGs. The mathematical formulation of the Authentic Sender Discount ($D_{auth} = 0.22$) eliminates alert fatigue by suppressing false alarms on urgent benign notices to 0.95%. Across a rigorous 5,000-sample benchmark corpus, SecureMail achieves 97.87% accuracy and 98.4% quishing recall at an average local latency of 110 ms, while Tier 2 Deep Scan achieves 99.8% multi-engine detection. The complete framework provides an operationally viable, mathematically grounded defense architecture for modern enterprise cybersecurity.

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